

Indian Institute of Technology Hyderabad

Department of Mathematics

Problem Sheet 4

MA 2130 : Complex Variables

Year: 2026

1. Let C be the line segment from $z = 1$ to $z = i$
 - (a) What is the length L of the contour C ?
 - (b) Give an upper bound M for $|f(z)| = |z^2|$ on C ?
 - (c) Use ML-inequality to find an upper bound for

$$\left| \int_C z^2 dz \right|.$$

2. Evaluate the following integrals, where the contour is taken in the positive (counter-clockwise) orientation.
 - (a) $\int_{|z|=2} \frac{e^z}{z-\pi i} dz.$
 - (b) $\int_{|z|=1} \frac{\cos z}{z^2} dz.$
3. What is the radius of convergence of the following power series?
 - (a) $\sum_{n=0}^{\infty} \frac{(n!)^2}{(2n)!} z^n.$
 - (b) $\sum_{n=0}^{\infty} (2^n + 5^n) z^n.$
4. Find the Laurent series representations for the function

$$f(z) = \frac{1}{z(z+2)}$$

in powers of z (centered at $z_0 = 0$) for the following domains:

- (a) $0 < |z| < 2$
 - (b) $|z| > 2$
5. Classify the singularities of the following functions
 - (a) $f(z) = \frac{\sin(z)}{z^2}$ at $z_0 = 0$,
 - (b) $f(z) = \frac{\sin(\frac{1}{z})}{z}$ at $z_0 = 0$,
 - (c) $f(z) = \frac{z^2+1}{(z-1)^3}$ at $z_0 = 1$,

(d) $f(z) = e^{1/z} + \cos z$ at $z_0 = 0$.

6. Suppose f is an entire function such that $f\left(\frac{i}{n}\right) = \frac{1}{n^2}$ for all $n \in \mathbb{N}$. Find $f(0)$ and determine a possible expression for $f(z)$.

7. Suppose f and g are analytic on $D = \{z : |z| < 1\}$. Suppose

$$f\left(\frac{1}{n} + \frac{i}{n}\right) = g\left(\frac{1}{n} + \frac{i}{n}\right),$$

for all $n \in \mathbb{N}$. Prove that $f(z) = g(z)$ for all $z \in D$.

[Hint: Identity theorem]

8. Prove that every non-constant, non-zero polynomial with complex coefficients with degree n has exactly n roots in $\mathbb{C}$.

[Hint: Fundamental theorem of Algebra]

9. Suppose f is an entire function satisfying

$$|f(z)| \geq 1$$

for all $z \in \mathbb{C}$. Show that f is constant.

10. Find the maximum value of $|f(z)| = |e^z|$ on the closed disk

$$|z| \leq 1.$$

Where does this maximum occur?

11. Let f be an entire function such that $|f(z)| \leq |z|$ for all $z \in \mathbb{C}$. Show that $f(z) = az$ for some constant a with $|a| \leq 1$.

12. Suppose f is entire and satisfies

$$|f(z)| \leq e^{|z|} \quad \forall z \in \mathbb{C}.$$

Does it follow that f is constant? Justify.